LAGOS BUSINESS SCHOOL · MEMBA 14 · EEGB · PART II

SESSION 8

Monetary Policy from QE to QT

From markets, to macro, to decisions that matter — in Nigeria and the world.



Reading: Note on Money and Monetary Policy (HBS 9-797-076) · Liquidity Management in Nigeria (CBN Brief) · Monetary Integration in West Africa (CBN Brief) · Facilitator: Elvis Ozegbe

Session 7 in Three Ideas

Real vs nominal

Most of the growth in naira GDP was inflation; real GDP grew far less.

Per person

Population growth ate almost all of the gain in income per person.

Rebasing

A new base changes levels and ratios, not what the economy did.

TODAY'S PLAN

Session 8 at a Glance — 75 Minutes

What central banks do, how a rate decision reaches your P&L, and why it takes so long.



Hook

From 18.75% to 27.5% in ten months

What Central Banks Do

Mandates, tools, QE and QT

Transmission & Nigeria

Channels, lags, liquidity, fiscal dominance

Lab

How fast does the rate reach your firm?

BY THE END OF TODAY

Session 8 Learning Objectives

You will be able to:

01

Explain the Toolkit

Describe what a central bank targets and how it uses the policy rate, reserves, open-market operations and its balance sheet (QE and QT).

02

Trace Transmission

Follow a rate decision through banks, markets, the exchange rate and expectations to prices and output — and explain the lags.

03

Judge the Stance

Use real rates, liquidity conditions and a simple rule to judge whether policy is tight or loose.

THE HOOK

Nearly Nine Points in Ten Months

“The MPC voted to raise the Monetary Policy Rate by 400 basis points to 22.75%.” — CBN, February 2024

Between July 2023 and November 2024 the CBN lifted its policy rate from 18.75% to 27.5%. For a business:

Borrowing

A ₦5bn overdraft at MPR + 6 cost about ₦450m a year more by end-2024 than in mid-2023.

Customers

Consumer and SME credit became scarce and dear; big-ticket purchases were postponed.

The naira

High naira yields began to attract portfolio dollars back — helping the exchange rate stabilise in 2025.

How Fast Does the CBN Reach Your Firm?

The situation

In 2024 the CBN raised its policy rate sharply to fight inflation of over 30%, tightened bank reserves and mopped up liquidity. Inflation stayed high for months. Firms asked whether the medicine was working.

The decision question

Through which **channels** does a rate rise reach a Nigerian firm, **how long** does it take, and was policy actually **tight**?

The data we have

ng_macro_monthly.csv — policy rate, interbank rate, inflation, money, credit, exchange rate and PMI, 2015–2025; global_policy_rates.csv — seven central banks, 2019–2025. *Stylised teaching data.*

What a good answer looks like

A short note for your CFO: what the 2024–25 rate path means for **funding costs, customer demand and the naira** over the next 12–18 months.

PART

01

What Central Banks Do

Mandates, tools and the balance sheet.

What Is a Central Bank For?

Price stability

The core job almost everywhere

- Keep inflation low and predictable
- Many banks set an explicit target or band
- Nigeria's CBN has long referred to a single-digit (6–9%) band
- Credibility makes the job easier

Other roles

Often added — sometimes in tension

- Financial-system stability and bank supervision
- Managing the exchange rate and reserves
- Banker to the government
- Development finance (CBN's past intervention schemes)

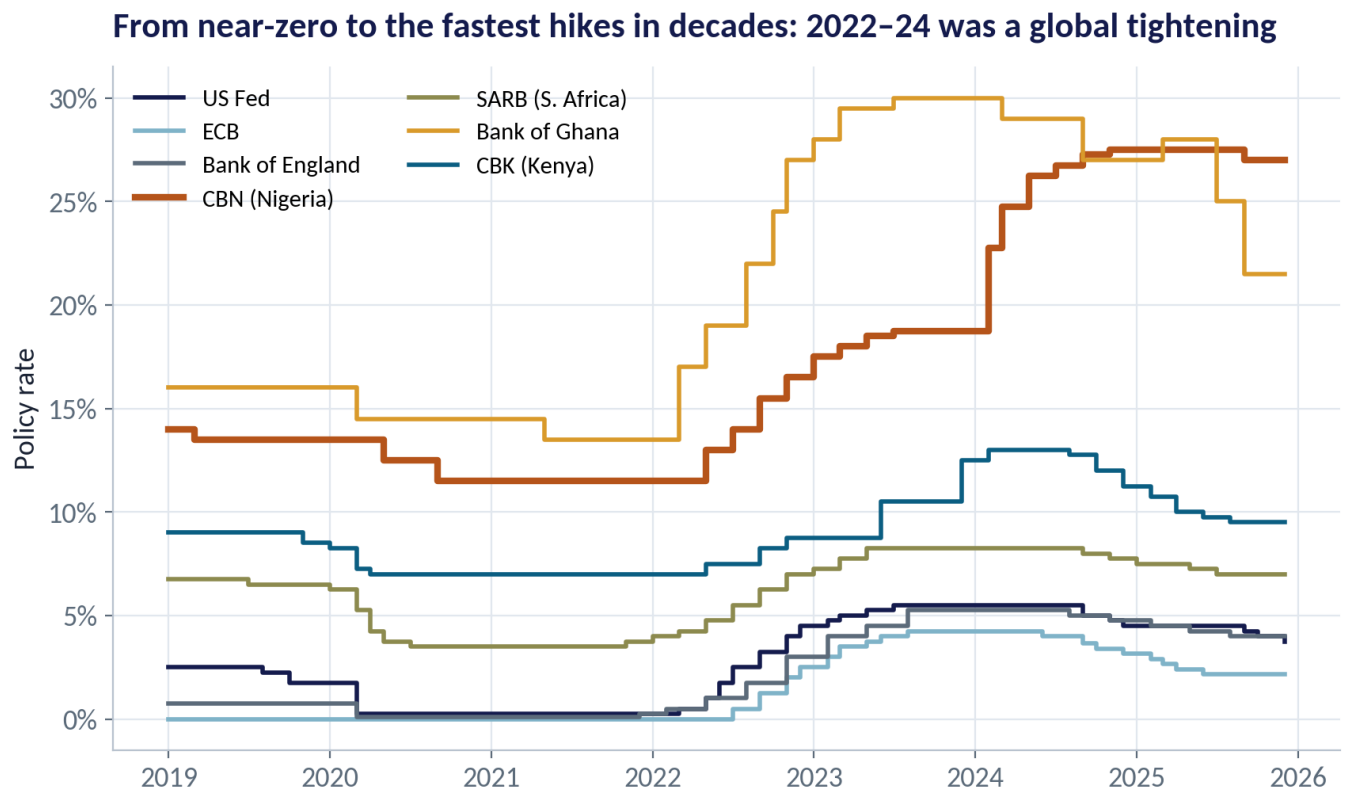
The more jobs a central bank takes on, the harder it is to hit any one of them — and the more its credibility on inflation suffers.

How the CBN Moves Money and Rates

Tool	What it does	Effect when tightened
Monetary policy rate (MPR)	Signals the price of central-bank money; anchors other rates	Loans, deposits and bond yields reprice upward
Standing facilities	Rates at which banks can borrow from or deposit with the CBN — a corridor around the MPR	Narrows where interbank rates can go
Cash reserve requirement (CRR)	Share of deposits banks must keep at the CBN, unremunerated	Less money for banks to lend
Open-market operations (OMO)	CBN sells or buys its own bills to drain or add liquidity	Drains naira; raises short-term yields
FX interventions	Selling or buying dollars from reserves	Supports or weakens the naira

In 2024 the CBN used several tools at once: large MPR hikes, a much higher CRR and heavy OMO sales.

A Worldwide Tightening Cycle



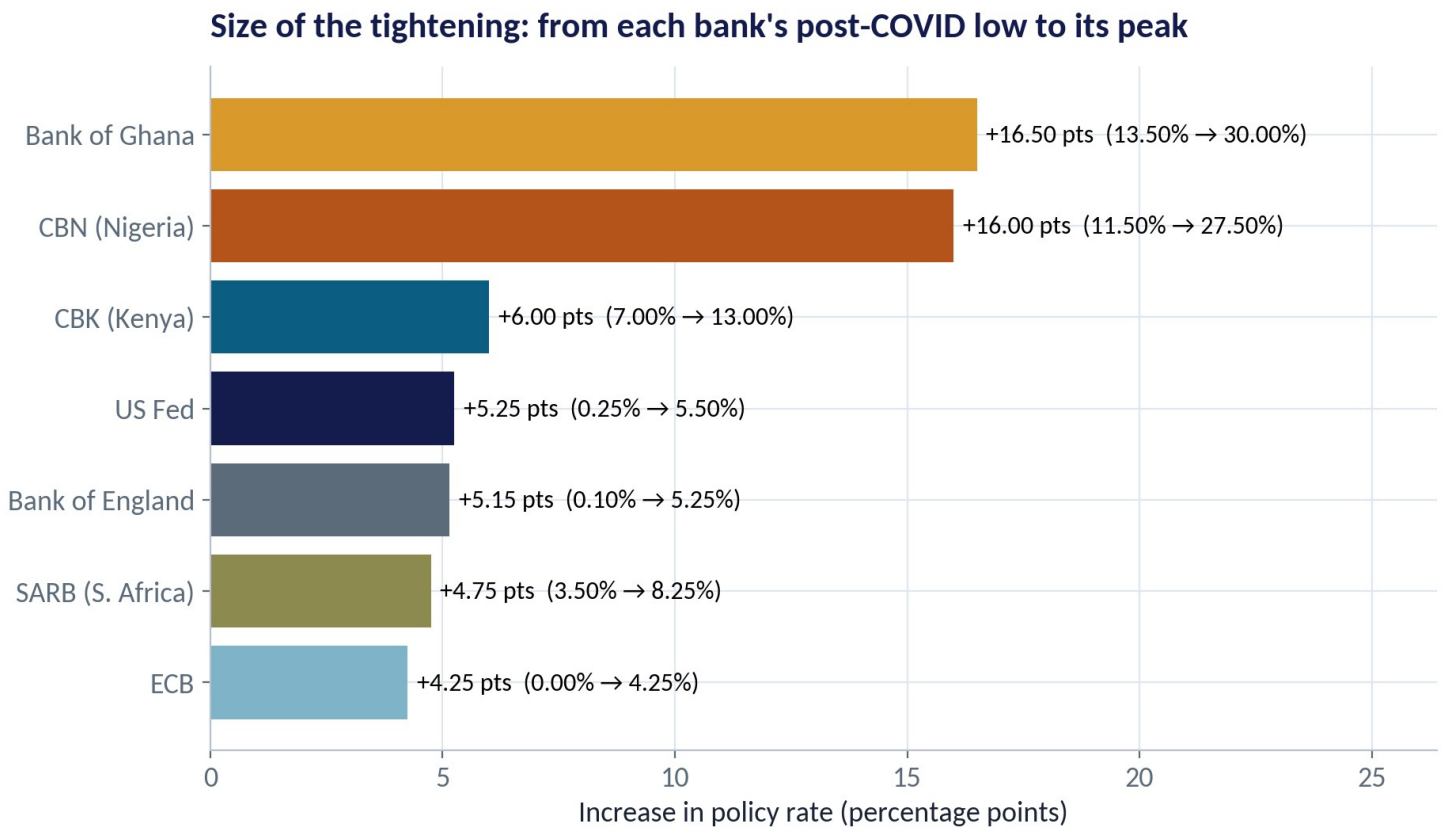
Source: EEGB stylised teaching data (data/). Not official statistics.

Fig S08-F01 · replicate: s08_figures.R / .py

WHAT TO SEE

- Most central banks cut to near zero in 2020 and kept rates there through 2021.
- In 2022–23 the US Fed, ECB and Bank of England raised rates at the fastest pace in decades.
- Higher US rates pulled dollars **out** of frontier markets like Nigeria and Ghana.
- From 2024–25 the rich-country banks began to cut; Nigeria stayed high.

Size of the Hiking Cycles



Source: EEGB stylised teaching data (data/). Not official statistics.

Fig S08-F02 · replicate: s08_figures.R / .py

WHAT TO SEE

- From their post-COVID lows to their peaks: Ghana **+16.5**, Nigeria **+16.0**, Kenya **+6.0** points.
- The US Fed raised by **5.25** points, from 0.25% to 5.5%.
- African banks hiked more because inflation and currency pressure were far higher.
- Bigger hikes also had to compensate investors for currency risk.

When the Rate Hits Zero: The Balance Sheet

IN PLAIN ENGLISH

Quantitative easing (QE): the central bank creates money to buy government bonds and other assets, pushing long-term rates down. **Quantitative tightening (QT)** is the reverse: it lets those bonds mature or sells them, draining money and pushing long rates up.

EXAMPLE

The US Fed's balance sheet grew to almost **$9 trillion** by 2022, then began shrinking under QT from mid-2022. Global dollar liquidity tightened just as Nigeria needed dollars most.

THE FORMULA

QE: CB buys bonds → reserves ↑, long yields ↓ · QT: reverse

WHY A MANAGER CARES

QE and QT in Washington change the cost and availability of dollars for Nigerian firms, banks and the government — through Eurobond yields and portfolio flows.

Quick Check: What Would You Say?

?

The CBN raises the cash reserve requirement while leaving the MPR unchanged. What is the most likely immediate effect?

A. Banks have more money to lend, so loan rates fall

B. Banks have less naira to lend or place in the interbank market, so liquidity tightens and short-term rates rise

C. The naira weakens because the CBN is printing money

D. Nothing — only the MPR matters

Discuss with your neighbour for 60 seconds — answer and reasoning in the speaker notes.

PART

02

How Policy Reaches Your Firm

Channels, lags and leaks.

From the MPC Meeting to Your P&L

Each link can be strong, weak or broken.



1. Policy rate

MPC raises the MPR; CBN drains liquidity

2. Market rates

Interbank, T-bills and bond yields rise

3. Banks

Loan and deposit rates reprice; credit is rationed

4. Naira & expectations

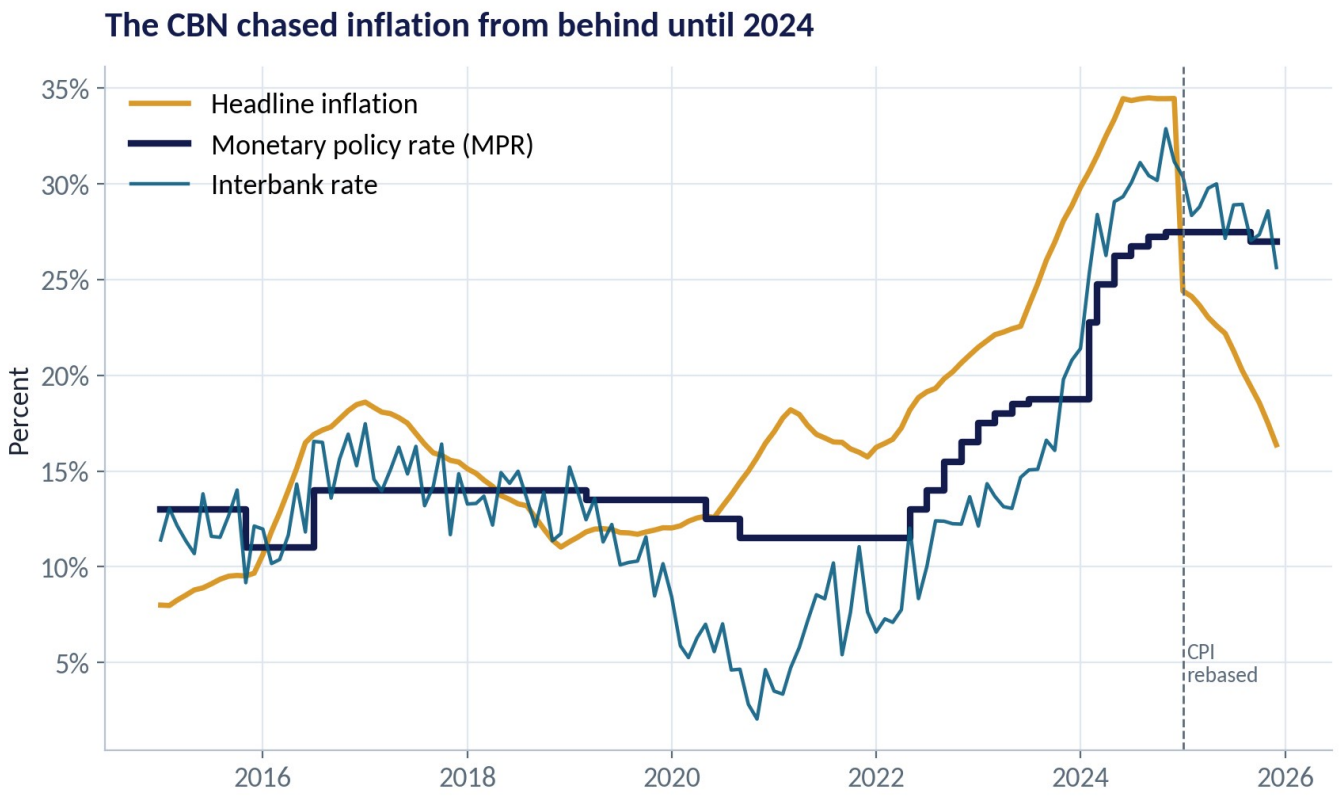
Higher yields attract dollars; credibility lowers expected inflation

5. Demand & prices

Spending and investment slow; price rises ease — after 12–24 months

*In Nigeria the **bank-lending** link is weak (credit is small relative to GDP) while the **exchange-rate** link is strong.*

Rates, Inflation and the Interbank Market



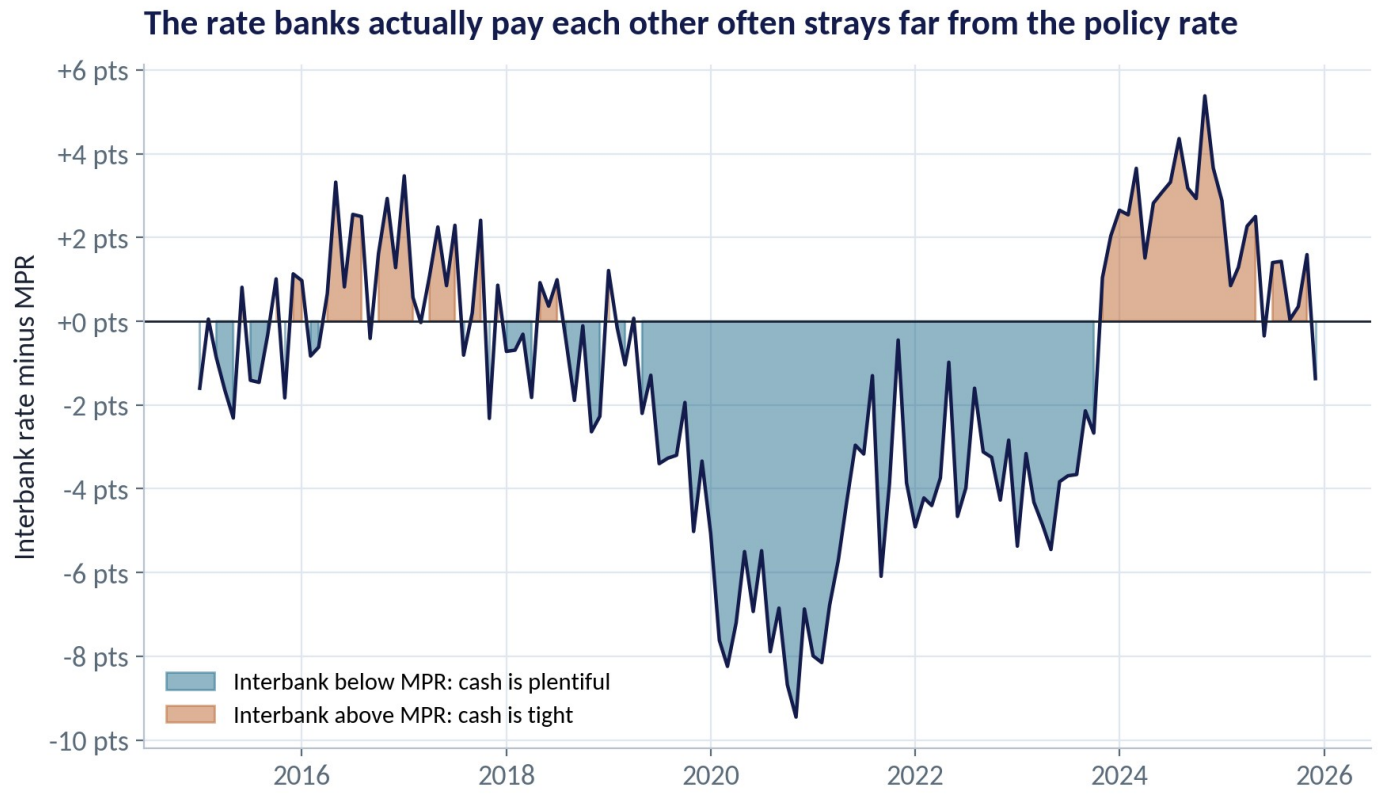
Source: EEGB stylised teaching data (data/). Not official statistics.

Fig S08-F03 · replicate: s08_figures.R / .py

WHAT TO SEE

- For most of 2020–2023 the MPR stayed **below** inflation — a negative real rate.
- The 2024 hikes closed much of the gap; in 2025 the real rate turned clearly positive (helped by the CPI rebasing).
- The **interbank** rate (teal) is what banks actually pay each other — it does not always follow the MPR.
- Judge policy by the **real** rate and the **interbank** rate, not the headline MPR alone.

When the Policy Rate Does Not Bind



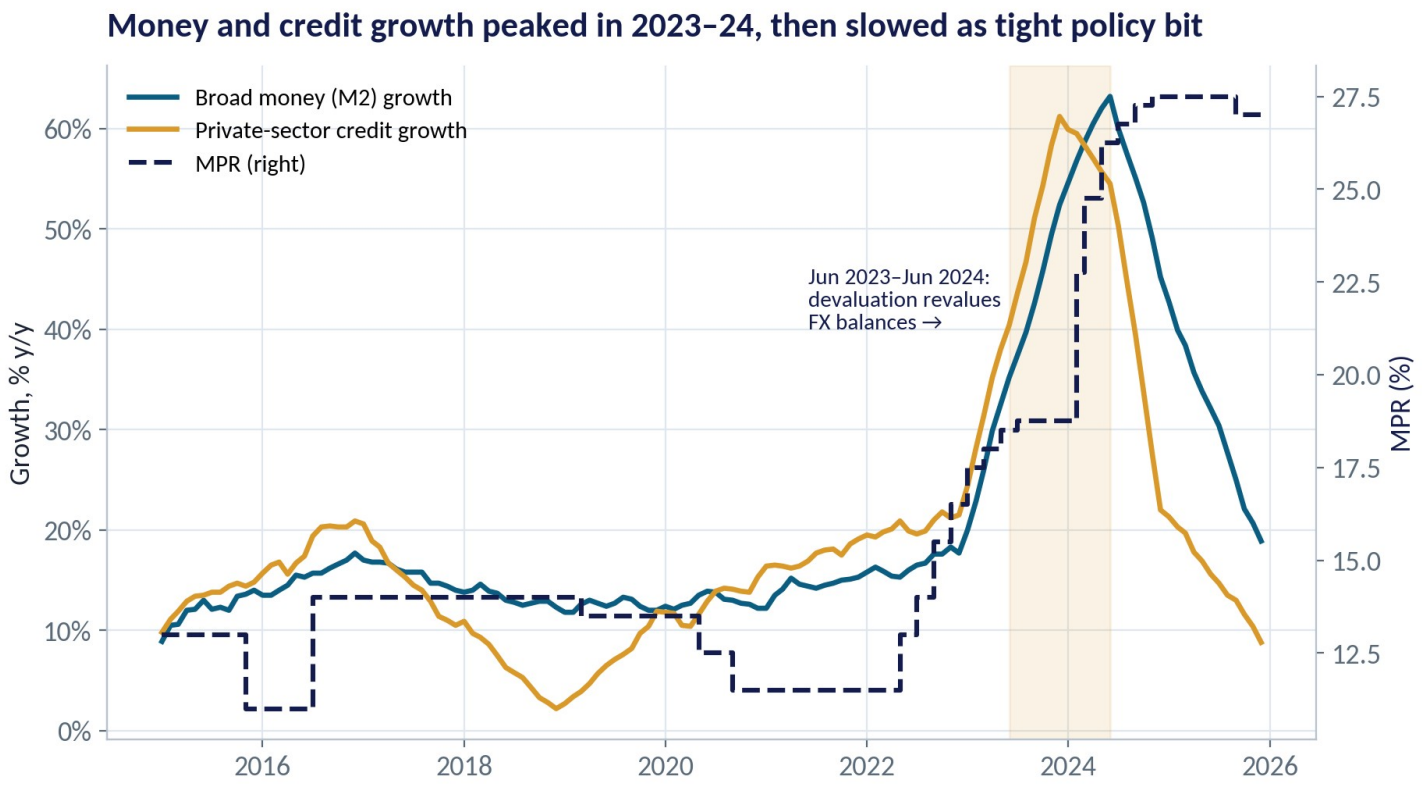
Source: EEGB stylised teaching data (data/). Not official statistics.

Fig S08-F04 · replicate: s08_figures.R / .py

WHAT TO SEE

- When banks are flush with naira, interbank rates fall far **below** the MPR — down to **−9.5 points** in late 2020.
- Then a 'high' MPR does little: money is cheap anyway.
- In 2024 the CBN drained liquidity and interbank rates rose **above** the MPR (up to about **+5.4 points**).
- That is when tightening really started to bite.

Did Tight Policy Slow Money and Credit?



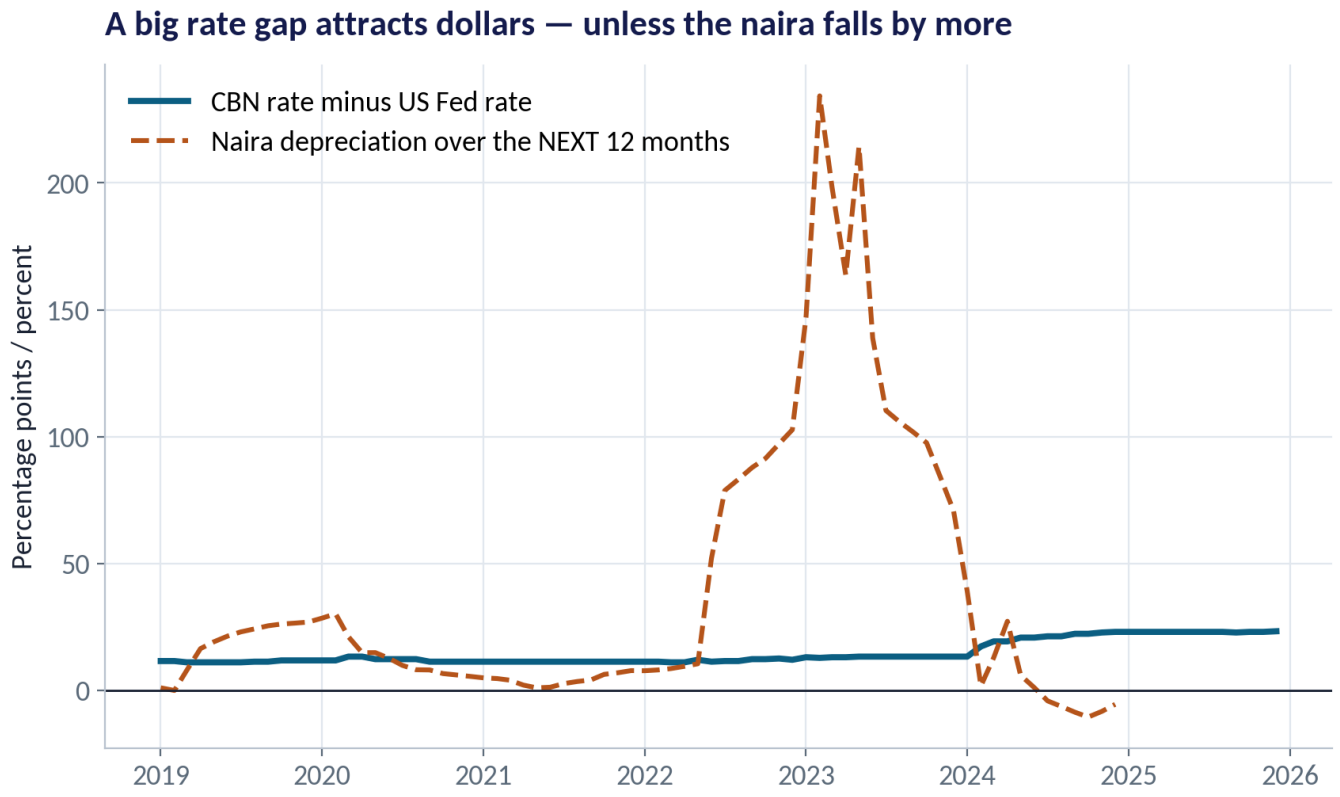
Source: EEGB stylised teaching data (data/). Not official statistics.

Fig S08-F05 · replicate: s08_figures.R / .py

WHAT TO SEE

- Credit growth peaked at about **61%** (Dec 2023) and broad money at about **63%** (Jun 2024).
- Much of that was **revaluation**: dollar loans and deposits were worth more naira after devaluation.
- As revaluation faded and policy tightened, growth fell to about **9%** (credit) and **19%** (M2) by end-2025.
- Lesson: in a devaluation, naira money and credit numbers overstate real expansion.

The Rate Gap and the Naira



Source: EEGB stylised teaching data (data/). Not official statistics.

Fig S08-F07 · replicate: s08_figures.R / .py

WHAT TO SEE

- The teal line is the gap between the CBN rate and the US Fed rate — what a dollar investor earns for holding naira.
- The dashed line is how much the naira **then** fell over the next 12 months.
- In 2022–23 the naira fell far more than the gap — investors lost money, so they stayed away.
- Once the gap exceeded expected depreciation in 2024–25, portfolio inflows returned.

How Managers Misread Monetary Policy

THE TRAP	THE BETTER HABIT
✗ "Rates went up last month, so inflation should fall this quarter."	✓ Expect **12–24 months** of lag; plan for high costs in the meantime.
✗ Reading the MPR as the cost of money.	✓ Watch the **interbank, T-bill and your own loan** rates.
✗ Ignoring the CRR and OMO because they are technical.	✓ They drain liquidity directly — often faster than the MPR.
✗ Assuming high rates always strengthen the naira.	✓ Only if investors expect the naira to fall **less** than the rate gap.
✗ Treating central-bank lending to government as harmless.	✓ Large monetary financing fuels inflation and undermines credibility (fiscal dominance).

PART

03

Judging the Stance

Tight, loose — and who is really in charge?

A RULE OF THUMB

The Taylor Rule

IN PLAIN ENGLISH

John Taylor proposed that a central bank should set its rate at a **neutral** level plus inflation, then add more when inflation is above target or the economy is running hot. It is a yardstick, not a law.

EXAMPLE

With inflation at **34.5%**, a 9% target, a 2% neutral real rate and a flat PMI, the rule gives about **49%** in December 2024 — far above the actual **27.5%**.

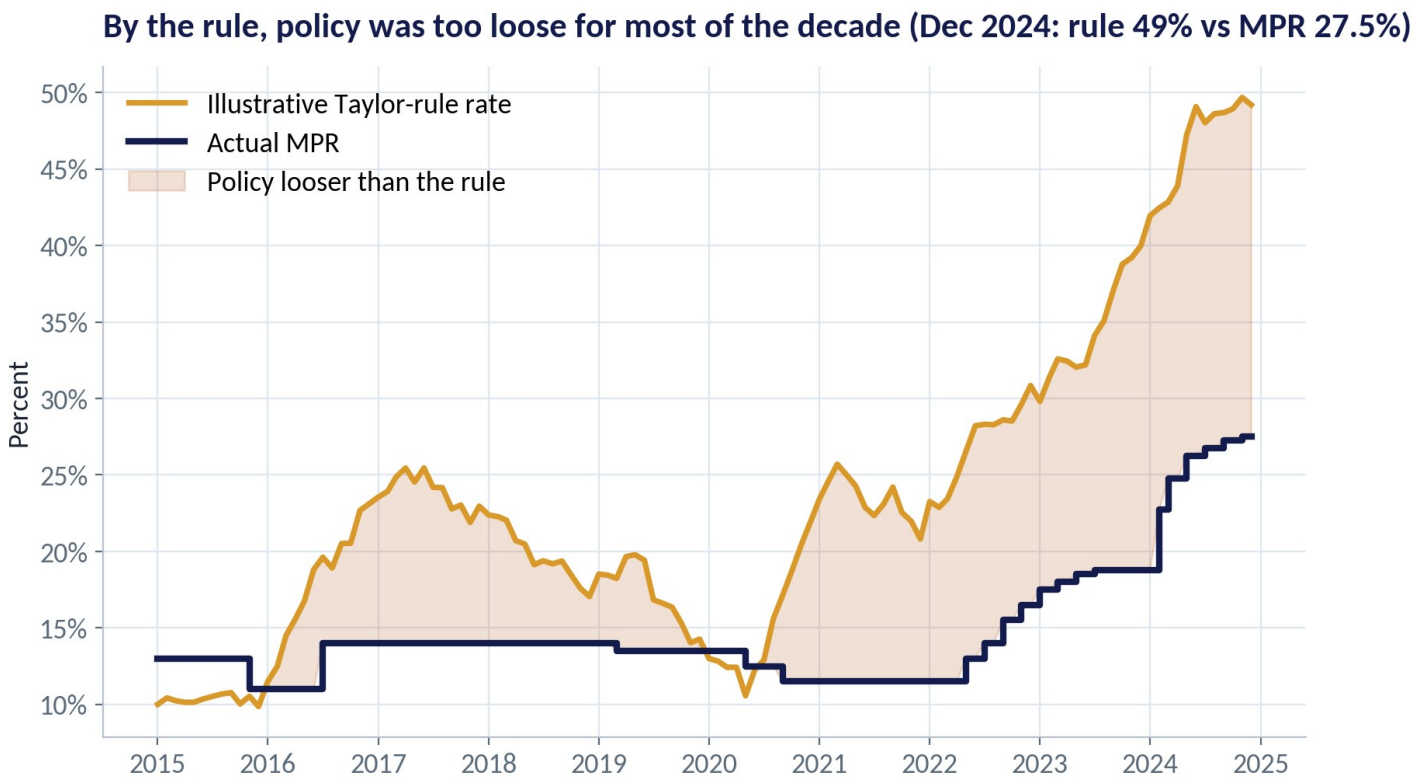
THE FORMULA

$$\text{rate} = r^* + \pi + 0.5(\pi - \text{target}) + 0.5 \times \text{output gap}$$

WHY A MANAGER CARES

The rule shows policy was loose by textbook standards. But against a **supply** shock, following it mechanically would crush output; the CBN chose to tighten hard but not that hard.

Was Policy Tight Enough?



Rule: rate = 2% + inflation + 0.5 × (inflation − 9%) + 0.5 × (PMI − 50). Illustrative assumptions; stylised data.

Fig S08-F06 · replicate: s08_figures.R / .py

WHAT TO SEE

- The gold line is the illustrative rule; navy is the actual MPR.
- The MPR sat **below** the rule in about **85%** of months, 2015–2024.
- The gap was widest in 2023–24, at the height of the inflation surge.
- Use the rule to ask 'how tight?', not to forecast the next MPC decision.

Fiscal Dominance

What it is

When government borrowing needs force the central bank to keep rates low or to lend to the government directly, inflation control comes second.

Nigeria's experience

Large CBN overdrafts to the federal government ('Ways and Means') built up before 2023 and were later converted into long-term bonds.

Why it matters to you

If markets believe the central bank serves the budget first, they expect more inflation — and price your loans and the naira accordingly.

Quick Check: What Would You Say?

?

Inflation is 30%, the MPR is 27.5% and the interbank rate is 31%. Which statement best describes the stance?

A. Very loose: the MPR is below inflation

B. Roughly neutral to tight in practice: the rate banks actually pay is above inflation, though the MPR itself is slightly below

C. Very tight: the MPR is above 25%

D. Impossible to judge without the Taylor rule

Discuss with your neighbour for 60 seconds — answer and reasoning in the speaker notes.

HANDS-ON LAB

Lab: How Fast Does the Rate Reach Your Firm?

YOU WILL NEED

Data

`data/ng_macro_monthly.csv`
`data/global_policy_rates.csv`

R script

`sessions/s08/R/s08_lab.R`

Python script

`sessions/s08/python/s08_lab.py`

15 minutes

1. List every CBN **policy-rate decision** and its size.
2. Compute the **real policy rate** by year.
3. Compare the **interbank rate** with the MPR to judge liquidity.
4. Correlate rate changes with **credit growth** at different lags — and ask why correlation misleads here.
5. Compute an illustrative **Taylor-rule** rate and compare it with the MPR.

Deliverable: a short note for your CFO on funding costs, customer demand and the naira over the next 12–18 months.

WORDS TO OWN

Today's Vocabulary in Plain English

Monetary policy rate	The CBN's benchmark rate signalling the price of money.
Interbank rate	The rate banks charge each other for overnight naira.
Cash reserve requirement	Share of deposits banks must hold at the CBN.
Open-market operations	Central-bank sales or purchases of securities to manage liquidity.
Quantitative easing	Central-bank asset purchases with newly created money.
Quantitative tightening	Shrinking the central bank's balance sheet.
Transmission	The chain from a policy decision to prices and output.
Real policy rate	Policy rate minus inflation.
Taylor rule	A formula for where the policy rate 'should' be.
Fiscal dominance	When government finance needs override inflation control.

KEY TAKEAWAYS

Session 8 in 5 Ideas

01 Central banks control the **price of money**; they use rates, reserves, OMO and their balance sheet.

02 **QE and QT** abroad change the cost of dollars for Nigeria.

03 Policy works through **banks, markets, the naira and expectations** — with long lags.

04 Judge the stance by **real** and **interbank** rates, not the headline MPR alone.

05 **Fiscal dominance** undermines credibility and raises everyone's borrowing costs.



NEXT › Session 9: Fiscal Policy & Debt

Before next session: Read 'Net Fiscal Operations' and the CBN brief on the Fiscal Responsibility Law. Find what share of federal revenue went on interest in the latest budget.